Chapter 3

Traditional Symmetric-Key Ciphers

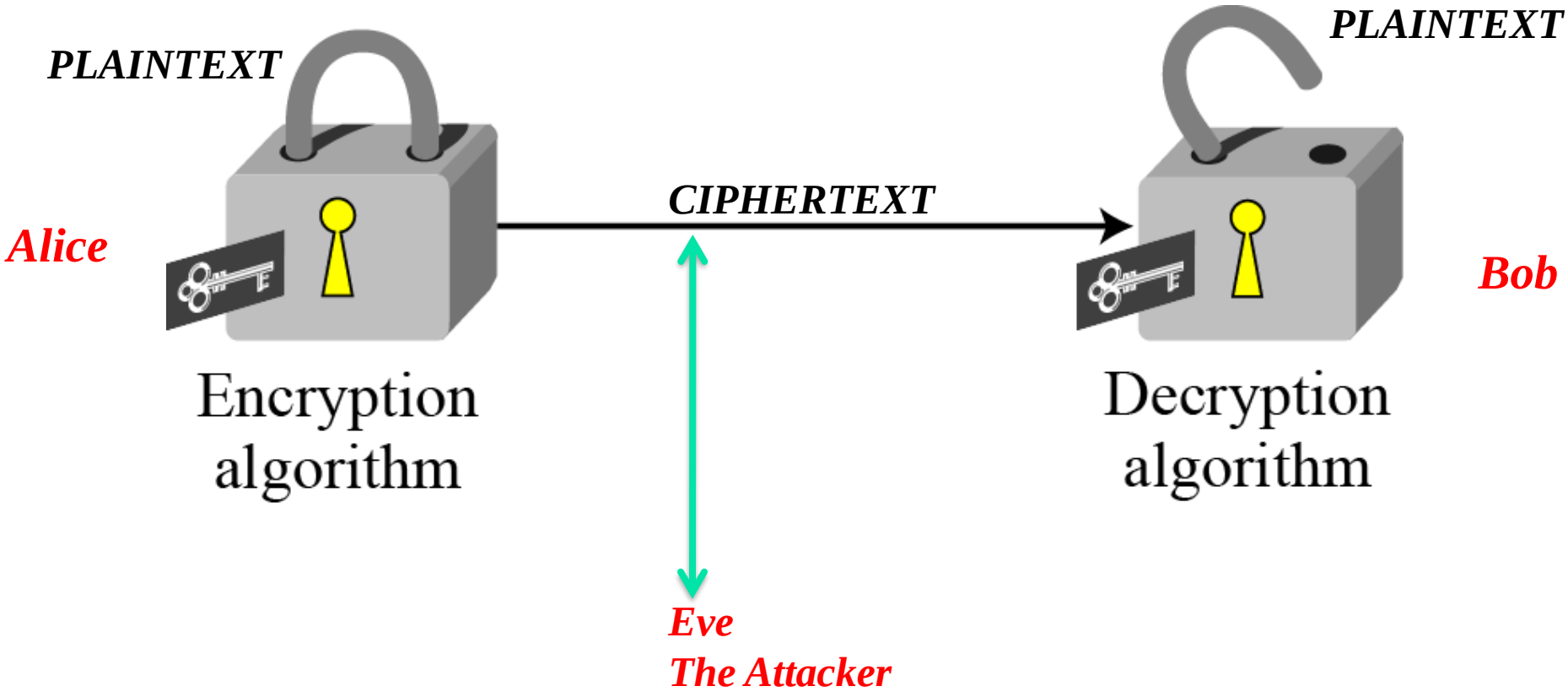
Chapter 3

Objectives

- To define the terms and the concepts of symmetric key ciphers
- To emphasize the two categories of traditional ciphers: substitution and transposition ciphers
- To describe the categories of cryptanalysis used to break the symmetric ciphers
- To introduce the concepts of the stream ciphers and block ciphers
- To discuss some very dominant ciphers used in the past, such as the Enigma machine

3.1 INTRODUCTION

Figure 3.2 *Locking and unlocking with the same key*

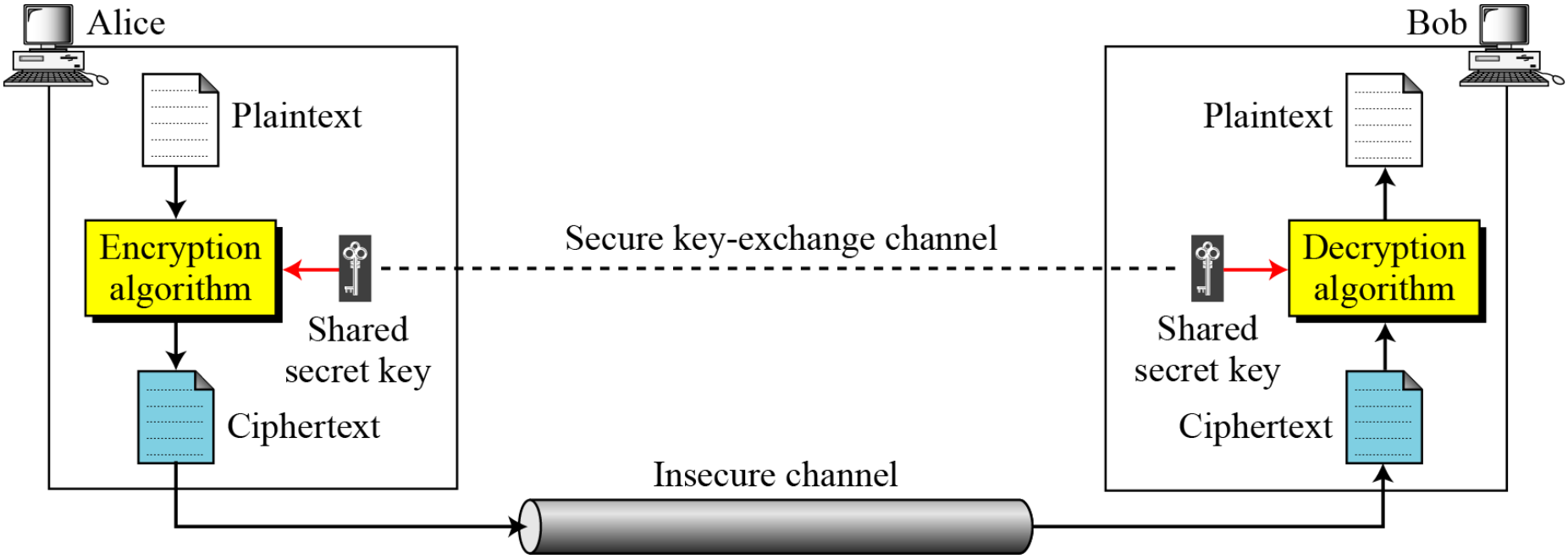


3-1 *Continued*

Components of Symmetric-key cipher:

1. ***The original message from Alice to Bob is called plaintext.***
2. ***The message that is sent through the channel is called the ciphertext.***
3. ***To create the ciphertext from the plaintext, Alice uses an encryption algorithm and a shared secret key.***
4. ***To create the plaintext from ciphertext, Bob uses a decryption algorithm and the same secret key.***
5. ***A shared secret key.***

Figure 3.1 *General idea of symmetric-key cipher*



3.1 Continued

If P is the plaintext, C is the ciphertext, and K is the key,

Encryption: $C = E_k(P)$ Decryption: $P = D_k(C)$

In which, $D_k(E_k(x)) = E_k(D_k(x)) = x$

We assume that Bob creates P_1; we prove that $P_1 = P$:

Alice: $C = E_k(P)$

Bob: $P_1 = D_k(C) = D_k(E_k(P)) = P$

3.1.1 Kerckhoff's Principle

Based on Kerckhoff's principle, one should always assume that the adversary, Eve, knows the encryption/decryption algorithm.

The resistance of the cipher to attack must be based only on the secrecy of the key.

3.1.2 Cryptanalysis

cryptography is the science and art of creating secret codes,
cryptanalysis is the science and art of breaking those codes.

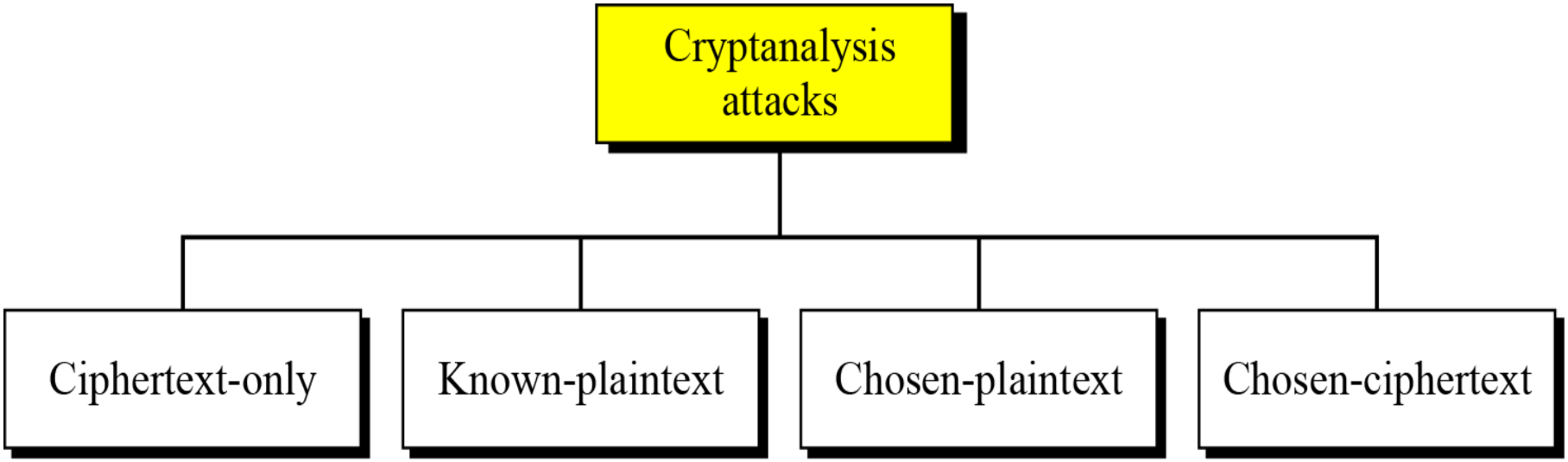
Cryptanalysis attacks

Statistical Attack : requires some statistical knowledge of the plaintext / language.

Brute-force Attack : try every possible keys.

3.1.2 Cryptanalysis

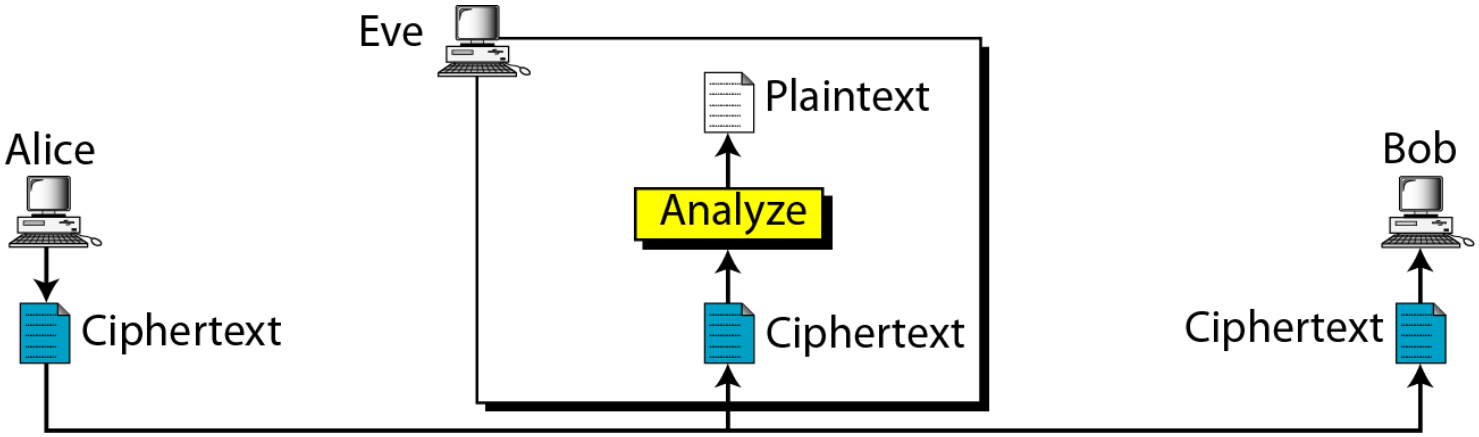
Figure 3.3 *Cryptanalysis attacks*



3.1.2 Continued

Ciphertext-Only Attack

Figure 3.4 *Ciphertext-only attack*



Known: only some ciphertext

Find: the key and the plaintext

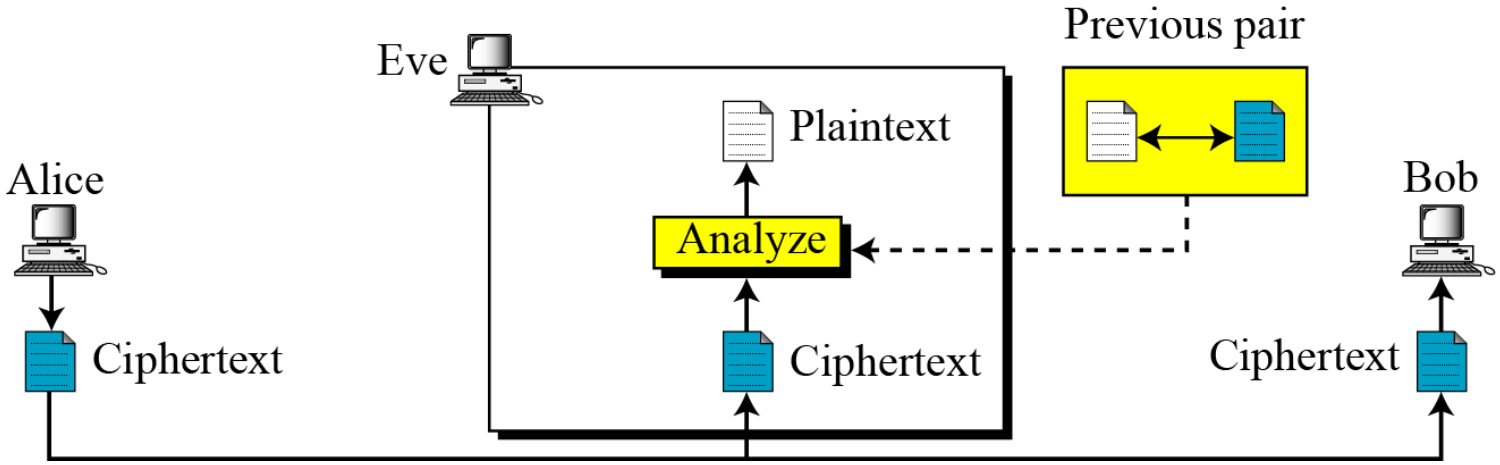
CT=UFYU , PT=?

Ans=TEXT

3.1.2 Continued

Known-Plaintext Attack

Figure 3.5 *Known-plaintext attack*



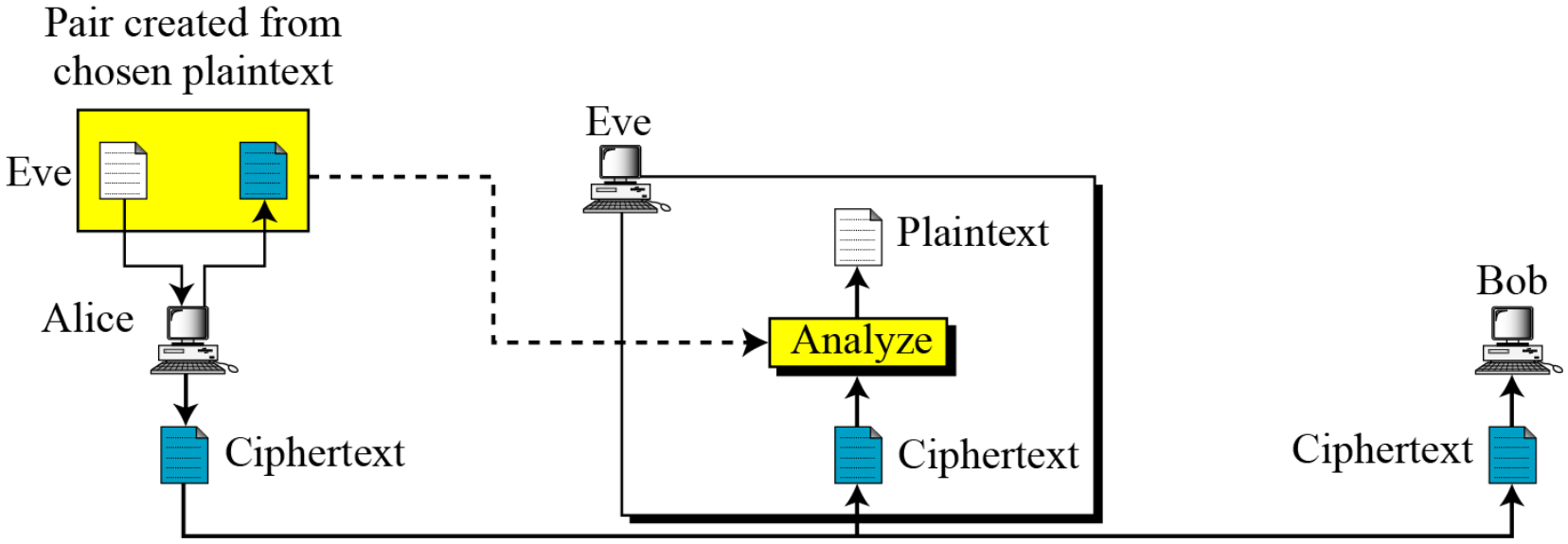
Known: a pair of plaintext-ciphertext and the intercepted ciphertext.

Find: the key and the plaintext

Ex: As SERUTAERC is to creatures so is ENOHPELET is to _____?

3.1.2 Continued

Chosen-Plaintext Attack (similar to Known-Plaintext Attack)



Known: a pair of plaintext-ciphertext but chosen by attacker herself and the intercepted ciphertext.

Find: the key and the plaintext

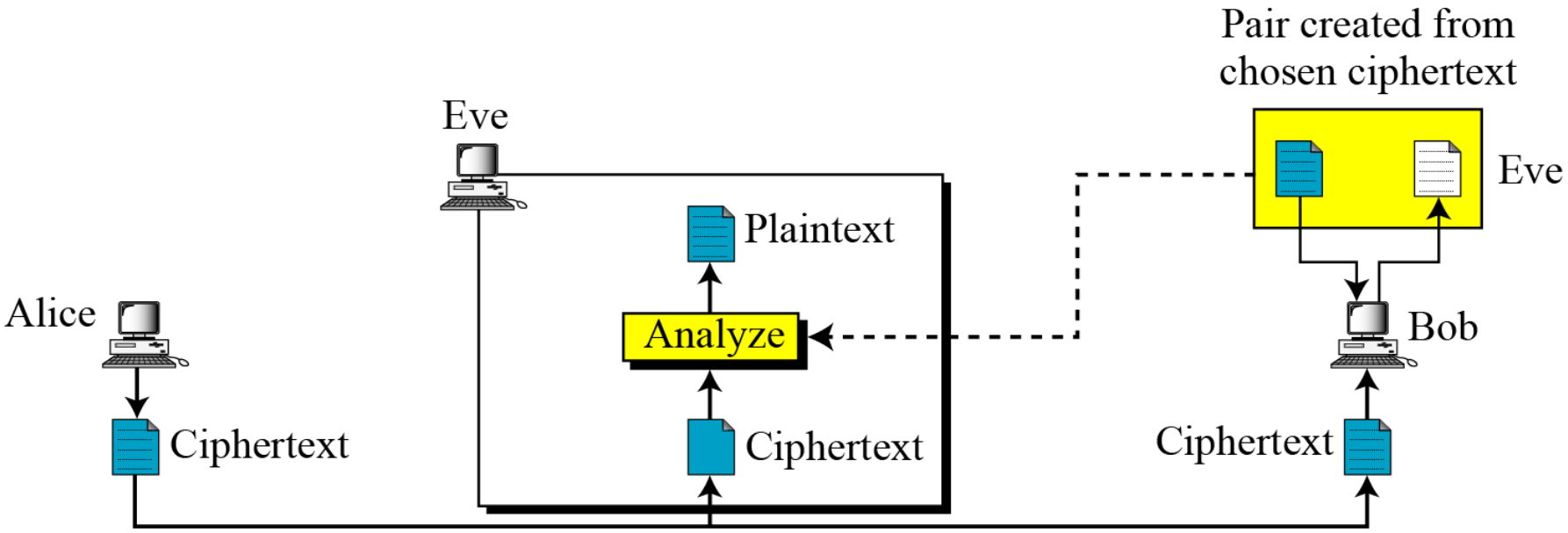
** Eve might have access to Alice's computer.*

Ex:If PT=PEREGRINATION and the CT=1232435678596

Given CT=244 PT=?

3.1.2 Continued

Chosen-Ciphertext Attack



Known: a pair of plaintext-ciphertext but chosen by attacker herself and the intercepted ciphertext.
Find: the key and the plaintext

Eve might have access to Bob's computer.

3-2 SUBSTITUTION CIPHERS

A substitution cipher replaces one symbol with another. Substitution ciphers can be categorized as either monoalphabetic ciphers or polyalphabetic ciphers.

Note

A substitution cipher replaces one symbol with another.

Topics discussed in this section:

3.2.1 Monoalphabetic Ciphres
3.2.2 Polyalphabetic Ciphers

3.2.1 Monoalphabetic Ciphers

Note

In monoalphabetic substitution, the relationship between a symbol in the plaintext to a symbol in the ciphertext is always one-to-one.

3.2.1 Continued

Example 3.1

The following shows a plaintext and its corresponding ciphertext. The cipher is probably monoalphabetic because both *l*'s (els) are encrypted as *O*'s.

Plaintext: hello **Ciphertext:** KHOOR

Example 3.2

The following shows a plaintext and its corresponding ciphertext. The cipher is not monoalphabetic because each *l* (el) is encrypted by a different character.

Plaintext: hello **Ciphertext:** ***ABNZF***

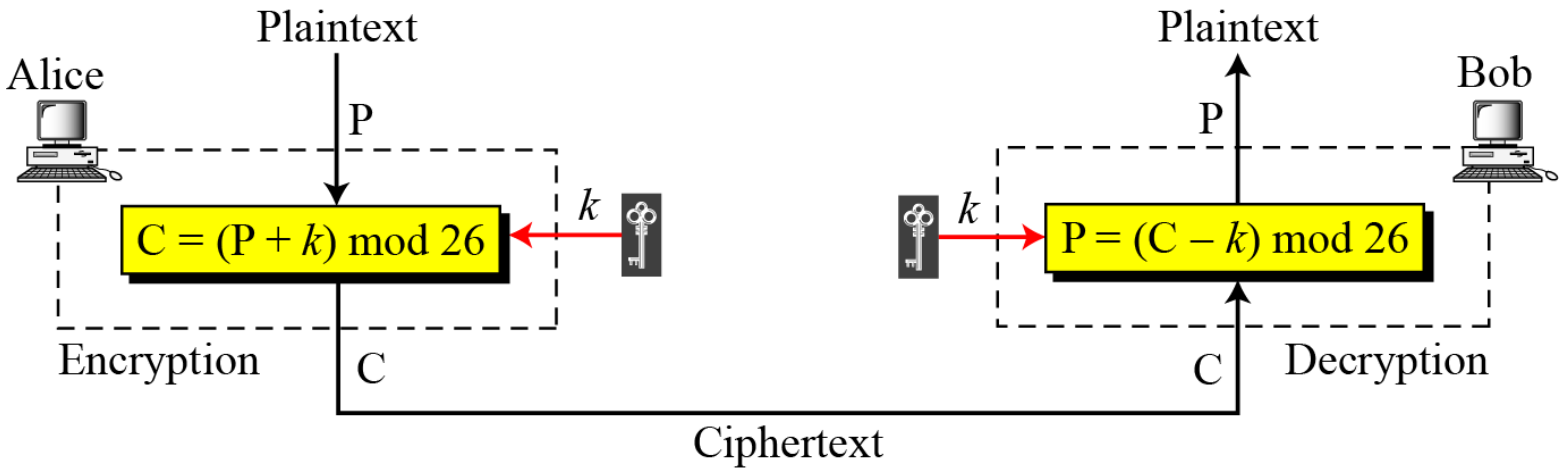
(Caesar cipher)
Additive Cipher

- **The simplest monoalphabetic cipher is the additive cipher.**
- **This cipher is sometimes called a shift cipher and sometimes a Caesar cipher, but the term additive cipher better reveals its mathematical nature.**

Figure 3.8 ***Plaintext and ciphertext in Z_{26}***

Plaintext →	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Ciphertext →	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Value →	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Figure 3.9 *Additive cipher*



Note

When the cipher is additive, the plaintext, ciphertext, and key are integers in Z_{26}.

Modular Arithmatic

In integer arithmetic,
if we divide a by n, we can get q and r .
The relationship between these four integers can be shown as

$a = q * n + r$ $\qquad$ $a \bmod n = r$

$11 \bmod 7 = 4$

$a = 11 \quad n = 7$

$11 = 1 \times 7 + 4$

$-11 \bmod 7 = ?$ 3

$-11 = -2 \times 7 + 3$

$-5 \bmod 2 = ?$

3.2.1 Continued

Example 3.3

Use the additive cipher with key = 15 to encrypt the message "hello".

Solution

We apply the encryption algorithm to the plaintext, character by character:

Plaintext: h → 07	Encryption: (07 + 15) mod 26	Ciphertext: 22 → W
Plaintext: e → 04	Encryption: (04 + 15) mod 26	Ciphertext: 19 → T
Plaintext: l → 11	Encryption: (11 + 15) mod 26	Ciphertext: 00 → A
Plaintext: l → 11	Encryption: (11 + 15) mod 26	Ciphertext: 00 → A
Plaintext: o → 14	Encryption: (14 + 15) mod 26	Ciphertext: 03 → D

Plaintext →	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Ciphertext →	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Value →	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

3.2.1 Continued

Example 3.4

Use the additive cipher with key = 15 to decrypt the message "WTAAD".

Solution

We apply the decryption algorithm to the plaintext character by character:

Ciphertext: W → 22	Decryption: (22 – 15) mod 26	Plaintext: 07 → h
Ciphertext: T → 19	Decryption: (19 – 15) mod 26	Plaintext: 04 → e
Ciphertext: A → 00	Decryption: (00 – 15) mod 26	Plaintext: 11 → l
Ciphertext: A → 00	Decryption: (00 – 15) mod 26	Plaintext: 11 → l
Ciphertext: D → 03	Decryption: (03 – 15) mod 26	Plaintext: 14 → o

Plaintext →	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Ciphertext →	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Value →	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

3.2.1 Continued

Shift Cipher and Caesar Cipher

- **Historically, additive ciphers are called shift ciphers.**
- **Julius Caesar used an additive cipher to communicate with his officers. For this reason, additive ciphers are sometimes referred to as the Caesar cipher.**
- **Caesar used a key of 3 for his communications.**

Note

Additive ciphers are sometimes referred to as shift ciphers or Caesar cipher.

3.2.1 Continued

Example 3.5

Eve has intercepted the ciphertext "UVACLYFZLJBYL". Show how she can use a brute-force attack to break the cipher.

Solution

Eve tries keys from 1 to 7.

With a key of 7, the plaintext is "not very secure", which makes sense.

Ciphertext: UVACLYFZLJBYL

K = 1	→	**Plaintext:** tuzbkxeykiaxk
K = 2	→	**Plaintext:** styajwdxjhzwj
K = 3	→	**Plaintext:** rsxzivcwigyvi
K = 4	→	**Plaintext:** qrwyhubvhfxuh
K = 5	→	**Plaintext:** pqvxgtaugewtg
K = 6	→	**Plaintext:** opuwfsztfdvsf
K = 7	→	**Plaintext:** notverysecure

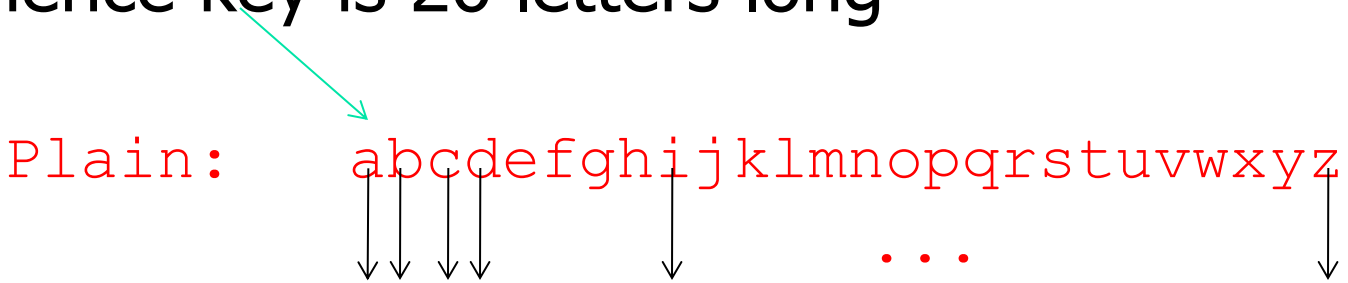


Cryptanalysis of Caesar Cipher

- only have 26 possible ciphers
  - A maps to A,B,..Z
- could simply try each in turn
- a **brute force search**
- given ciphertext, just try all shifts of letters
- do need to recognize when have plaintext
- eg. break the ciphertext "GCUA VQ DTGCM"
- Ans: P.T.- easy to break (key=2)

Continued…

- rather than just shifting the alphabet
- could shuffle (jumble) the letters arbitrarily
- each plaintext letter maps to a different random ciphertext letter
- hence key is 26 letters long



Plaintext: ifwewishtoreplaceletters

Ciphertext: WIRFRWAJUHYFTSDVFSFUUFYA



Continued…

- now have a total of $26! = 4 \times 10^{26}$ keys
- with so many keys, might think is secure
- but would be **!!!WRONG!!!**
- The problem is language characteristics
- human languages are redundant

  •in English e is by far the most common letter then T,R,N,I,O,A,S

3.2.1 Continued

Table 3.1 Frequency of characters in English

Letter	Frequency	Letter	Frequency	Letter	Frequency	Letter	Frequency
E	12.7	H	6.1	W	2.3	K	0.08
T	9.1	R	6.0	F	2.2	J	0.02
A	8.2	D	4.3	G	2.0	Q	0.01
O	7.5	L	4.0	Y	2.0	X	0.01
I	7.0	C	2.8	P	1.9	Z	0.01
N	6.7	U	2.8	B	1.5		
S	6.3	M	2.4	V	1.0		

Table 3.2 Frequency of diagrams and trigrams

Digram	TH, HE, IN, ER, AN, RE, ED, ON, ES, ST, EN, AT, TO, NT, HA, ND, OU, EA, NG, AS, OR, TI, IS, ET, IT, AR, TE, SE, HI, OF
Trigram	THE, ING, AND, HER, ERE, ENT, THA, NTH, WAS, ETH, FOR, DTH

3.2.1 Continued

Example 3.6

Eve has intercepted the following ciphertext. Using a statistical attack, find the plaintext.

XLILSYWIMWRSAJSVWEPIJSVJSYVQMPPMSRHSPPEVWMXMWASVX-LQSVILY-VVCFIJSVIXLIWIPPIVVIGIMZIWQSVISJJIVW

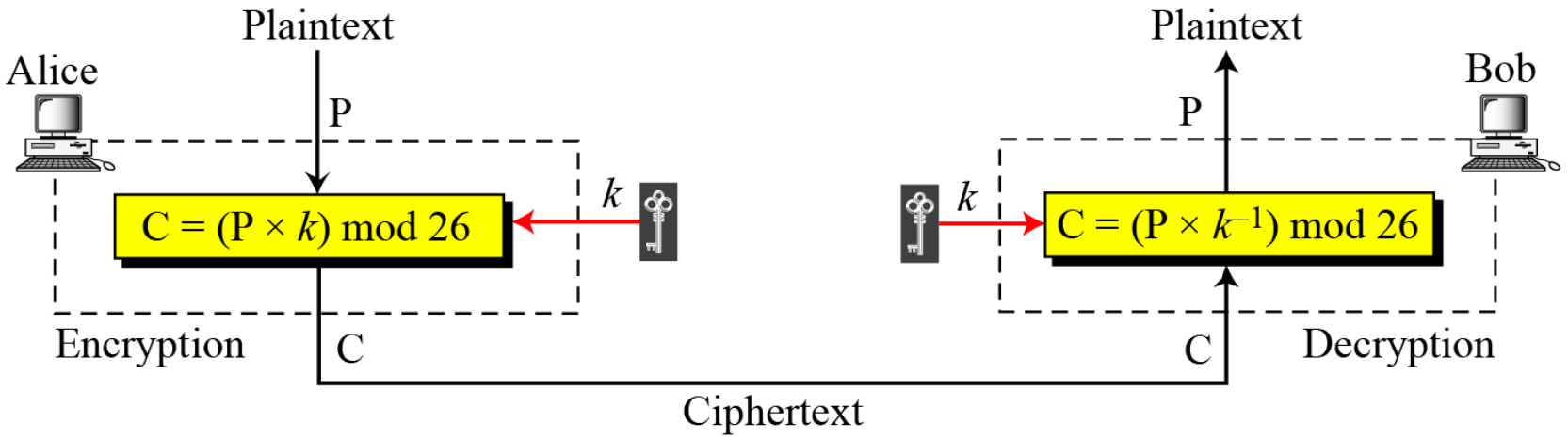
Solution

When Eve tabulates the frequency of letters in this ciphertext, she gets: I =14, V =13, S =12, and so on. The most common character is I with 14 occurrences. This means key = 4.

the house is now for sale for four million dollars it is worth more hurry before the seller receives more offers

Multiplicative Ciphers

Figure 3.10 *Multiplicative cipher*



Note

In a multiplicative cipher, the plaintext and ciphertext are integers in Z_{26}; the key is an integer in Z_{26}*.

3.2.1 Continued

Example 3.7

What is the key domain for any multiplicative cipher?

Solution

The key needs to be in Z_{26}^*. This set has only 12 members: 1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25.

Example 3.8

We use a multiplicative cipher to encrypt the message "hello" with a key of 7. The ciphertext is "XCZZU".

Plaintext: h → 07	Encryption: $(07 \times 07) \bmod 26$	ciphertext: 23 → X
Plaintext: e → 04	Encryption: $(04 \times 07) \bmod 26$	ciphertext: 02 → C
Plaintext: l → 11	Encryption: $(11 \times 07) \bmod 26$	ciphertext: 25 → Z
Plaintext: l → 11	Encryption: $(11 \times 07) \bmod 26$	ciphertext: 25 → Z
Plaintext: o → 14	Encryption: $(14 \times 07) \bmod 26$	ciphertext: 20 → U

For Decryption use the multiplicative inverse modulo of 7 i.e. $7^{-1} = 15$

3.2.2 Polyalphabetic Ciphers

In polyalphabetic substitution, each occurrence of a character may have a different substitute. The relationship between a character in the plaintext to a character in the ciphertext is one-to-many.

Autokey Cipher

$$P = P_1 P_2 P_3 \ldots \qquad C = C_1 C_2 C_3 \ldots \qquad k = (k_1, P_1, P_2, \ldots)$$

Encryption: $C_i = (P_i + k_i) \bmod 26$ Decryption: $P_i = (C_i - k_i) \bmod 26$

3.2.2 Continued

Example 3.14

Assume that Alice and Bob agreed to use an autokey cipher with initial key value $k_1 = 12$. Now Alice wants to send Bob the message "Attack is today". Enciphering is done character by character.

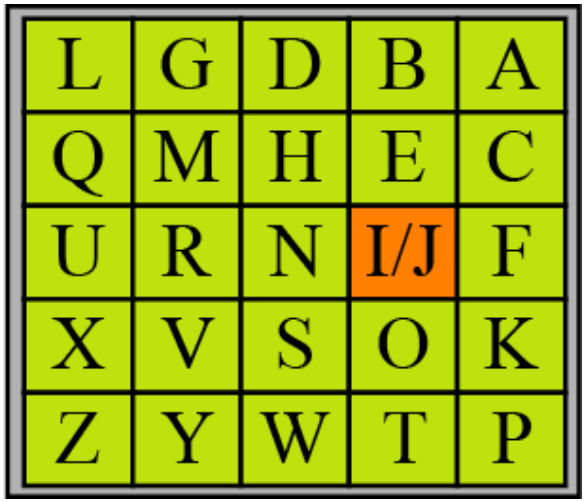
Plaintext:	a	t	t	a	c	k	i	s	t	o	d	a	y
P's Values:	00	19	19	00	02	10	08	18	19	14	03	00	24
Key stream:	*12*	*00*	*19*	*19*	*00*	*02*	*10*	*08*	*18*	*19*	*14*	*03*	*00*
C's Values:	12	19	12	19	02	12	18	00	11	7	17	03	24
Ciphertext:	**M**	**T**	**M**	**T**	**C**	**M**	**S**	**A**	**L**	**H**	**R**	**D**	**Y**

3.2.2 Continued

Playfair Cipher

Figure 3.13 *An example of a secret key in the Playfair cipher*

Secret Key =



L	G	D	B	A
Q	M	H	E	C
U	R	N	I/J	F
X	V	S	O	K
Z	Y	W	T	P

Example 3.15

Let us encrypt the plaintext "hello" using the key in Figure 3.13.

he → EC lx → QZ lo → BX

Plaintext: hello Ciphertext: ECQZBX

3.2.2 Playfair Cipher:

- *a 5X5 matrix of letters based on a keyword*
- *fill in letters of keyword (sans duplicates)*
- *fill rest of matrix with other letters*
- *eg. using the keyword COMPATIBLE*.

C	O	M	P	A
T	I/J	B	L	E
D	F	G	H	K
N	Q	R	S	U
V	W	X	Y	Z

Encrypting and Decrypting

- plaintext encrypted two letters at a time:

1. if a pair is a repeated letter, insert a filler like 'X', eg. "balloon" encrypts as "ba lx lo on"
2. if both letters fall in the same row, replace each with letter to right (wrapping back to start from end), eg. “ar" encrypts as "RM"
3. if both letters fall in the same column, replace each with the letter below it (again wrapping to top from bottom), eg. “mu" encrypts to "CM"
4. otherwise each letter is replaced by the one in its row in the column of the other letter of the pair, eg. “hs" encrypts to "BP", and “ea" to "IM" or "JM" (as desired)

Ex:

- Given keyword
  "simple"

s	i/j	m	p	l
e	a	b	c	d
f	g	h	k	n
o	q	r	t	u
v	w	x	y	z

- PT=Balloon, CT= ??
- PT- we are discovered save yourself

- CT- vabqaemietsfobelewcvqoomdshv



Security of the Playfair Cipher

- security much improved over monoalphabetic
- since have 26 x 26 = 676 digrams
- would need a 676 entry frequency table to analyse (verses 26 for a monoalphabetic)
- and correspondingly more ciphertext
- was widely used for many years (eg. US & British military in WW1)
- it **can** be broken, given a few hundred letters
- since still has much of plaintext structure

Vigenere Cipher

$P = P_1P_2P_3 \ldots$ $C = C_1C_2C_3 \ldots$ $K = [(k_1, k_2, \ldots, k_m), (k_1, k_2, \ldots, k_m), \ldots]$

Encryption: $C_i = P_i + k_i$ Decryption: $P_i = C_i - k_i$

Example 3.16

We can encrypt the message "She is listening" using the 6-character keyword "PASCAL".

Plaintext:	s	h	e	i	s	l	i	s	t	e	n	i	n	g
P's values:	18	07	04	08	18	11	08	18	19	04	13	08	13	06
Key stream:	***15***	***00***	***18***	***02***	***00***	***11***	***15***	***00***	***18***	***02***	***00***	***11***	***15***	***00***
C's values:	**07**	**07**	**22**	**10**	**18**	**22**	**23**	**18**	**11**	**6**	**13**	**19**	**02**	**06**
Ciphertext:	**H**	**H**	**W**	**K**	**S**	**W**	**X**	**S**	**L**	**G**	**N**	**T**	**C**	**G**

3.2.2 Continued

Example 3.16

- Let us see how we can encrypt the message "She is listening" using the 6-character keyword "PASCAL".
- The initial key stream is (15, 0, 18, 2, 0, 11).
- The key stream is the repetition of this initial key stream (as many times as needed).

i.e. P A S C A L
15, 0, 18, 2, 0, 11

Plaintext:	s	h	e	i	s	l	i	s	t	e	n	i	n	g
P's values:	18	07	04	08	18	11	08	18	19	04	13	08	13	06
Key stream:	*15*	*00*	*18*	*02*	*00*	*11*	*15*	*00*	*18*	*02*	*00*	*11*	*15*	*00*
C's values:	**07**	**07**	**22**	**10**	**18**	**22**	**23**	**18**	**11**	**6**	**13**	**19**	**02**	**06**
Ciphertext:	**H**	**H**	**W**	**K**	**S**	**W**	**X**	**S**	**L**	**G**	**N**	**T**	**C**	**G**

this additive cipher is a special case of Vigenere cipher. Where m=1

plaintext →

key ↓

	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	v	v	w	x	y	z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Table 3.3

A Vigenere Tableau

3.2.2 Continued

Vigenere Cipher (Crypanalysis)

Example 3.19

Let us assume we have intercepted the following ciphertext:

```
LIOMWGFEGGDVWGHHCQUCRHRWAGWIOWQLKGZETKKMEVLWPCZVGTH-
VTSGXQOVGCSVETQLTJSUMVWVEUVLXEWSLGFZMVVWLGYHCUSWXQH-
KVGSHEEVFLCFDGVSUMPHKIRZDMPHHBVWVWJWIXGFWLTSHGJOUEEHH-
VUCFVGOWICQLTJSUXGLW
```

The Kasiski test for repetition of three-character segments yields the results shown in Table 3.4.

String	*First Index*	*Second Index*	*Difference*
JSU	68	168	100
SUM	69	117	48
VWV	72	132	60
MPH	119	127	8

3.2.2 Continued

The greatest common divisor of differences is 4, which means that the key length is multiple of 4. First try $m = 4$ with frequency analysis.

```
C1: LWGWCRAOKTEPGTQCTJVUEGVGUQGECVPRPVJGTJEUGCJG
P1: jueuapymircneroarhtsthihytrahcieixsthcarrehe
C2: IGGGQHGWGKVCTSOSQSWVWFVYSHSVFSHZHWWFSOHCOQSL
P2: usssctsiswhofeaeceihcetesoecatnpntherhctecex
C3: OFDHURWQZKLZHGVVLUVLSZWHWKHFDUKDHVIWHUHFWLUW
P3: lcaerotnwhiwedssirsiirhketehretltiideatrairt
C4: MEVHCWILEMWVVXGETMEXLMLCXVELGMIMBWXLGEVVITX
P4: iardysehaisrrtcapiafpwtethecarhaesfterectpt
```

In this case, the plaintext makes sense.

Julius Caesar used a cryptosystem in his wars, which is now referred to as Caesar cipher. It is an additive cipher with the key set to three. Each character in the plaintext is shifted three characters to create ciphertext.

Example

- eg using keyword *deceptive*

```
key:        deceptivedeceptivedeceptive
plaintext:  wearediscoveredsaveyourself
ciphertext: ZICVTWQNGRZGVTWAVZHCQYGLMGJ
```

string	1st index	2nd index	difference
VTW	4	13	9

- ***suggests keyword size of 3 or 9***
- ***then attack each monoalphabetic cipher individually using previous techniques***

3.2.2 Continued

One-Time Pad

- *if a truly random key as long as the message is used, the cipher will be secure*
  - *called a One-Time pad*
  - *is unbreakable since ciphertext bears no statistical relationship to the plaintext*
  - *since for any plaintext & any ciphertext there exists a key mapping one to other*
  - *can only use the key once though*
- *have problem of safe distribution of key*

Enigma Machine

- **Enigma** was a portable cipher machine used to encrypt and decrypt secret messages.
  - a family of related electro-mechanical rotor machines



German military



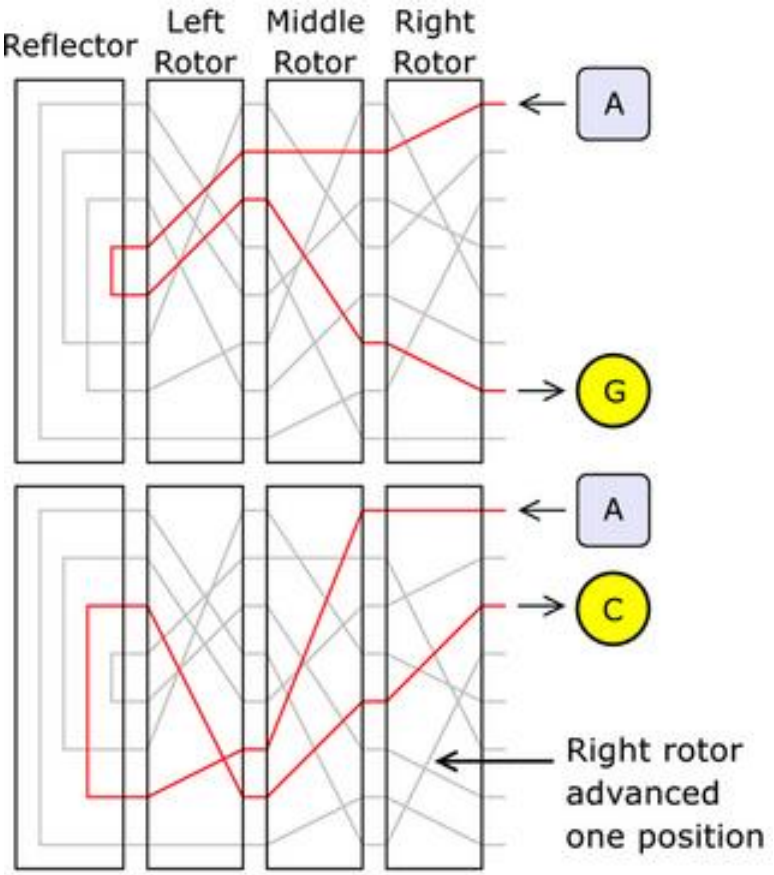
Japan commercial

Enigma Machine

Enigma encryption for two consecutive letters —

current is passed into set of rotors, around the reflector, and back out through the rotors again.

Letter A encrypts differently with consecutive key presses, first to G, and then to C. This is because the right hand rotor has stepped, sending the signal on a completely different route.



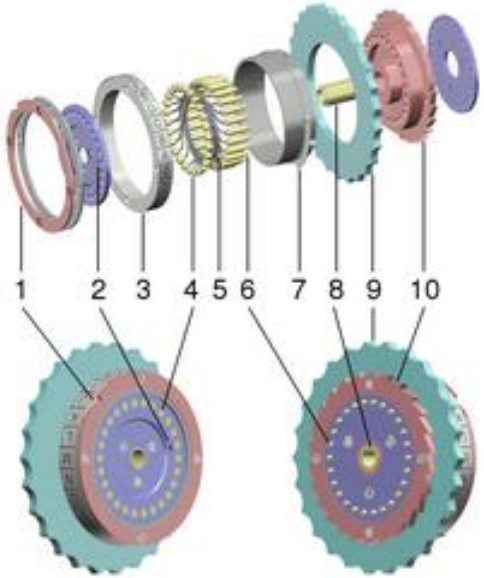
Enigma

- the actual encipherment of a letter is performed electrically.
  - When a key is pressed, the circuit is completed; current flows through the various components and ultimately lights one of many lamps, indicating the output letter.
  - Current flows from a battery through the switch controlled by the depressed key into a fixed *entry wheel*. This leads into the rotor assembly (or *scrambler*), where the complex internal wiring of each rotor results in the current passing from one rotor to the next along a convoluted path. After passing through all the rotors, current enters the *reflector*, which relays the signal back out again through the rotors and the entry wheel — this time via a different path — and, finally, to one of the lamps (the earliest Enigma models do not have the reflector).

Rotors

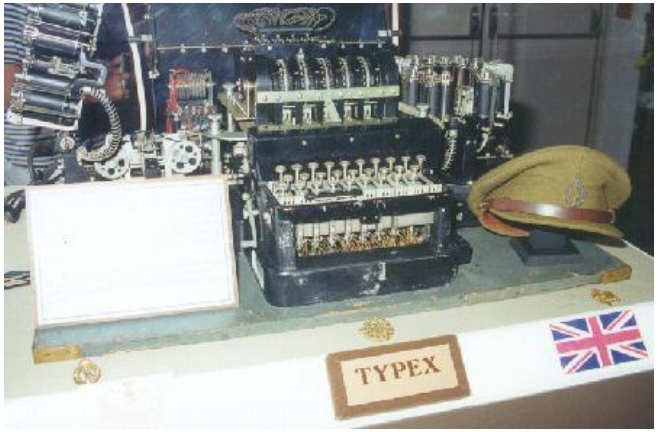
- performs a very simple type of encryption
  - a simple substitution cipher





World War II Era Encryption Devices





- A few here
  - Sigaba (*United States*)
  - Typex (*Britain*)
  - Lorenz cipher (*Germany*)





- For more, see
  - http://w1tp.com/enigma/

3-3 TRANSPOSITION CIPHERS

A transposition cipher does not substitute one symbol for another, instead it changes the location of the symbols.

Note

A transposition cipher reorders symbols.

- *these hide the message by rearranging the letter order without altering the actual letters used.*

Topics discussed in this section:

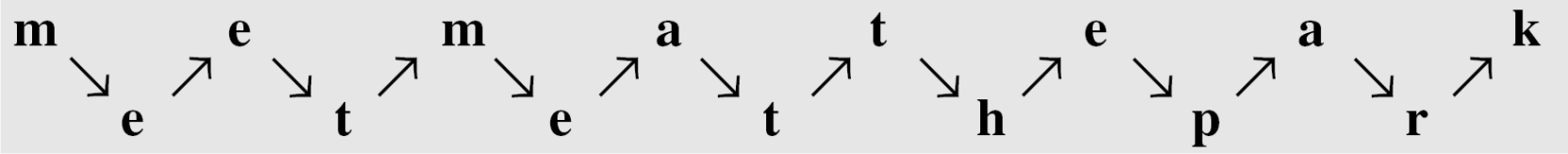
3.3.1 Keyless Transposition Ciphers
3.3.2 Keyed Transposition Ciphers
3.3.3 Combining Two Approaches

3.3.1 Keyless Transposition Ciphers

Simple transposition ciphers, which were used in the past, are keyless.

Example 3.22

A good example of a keyless cipher using the first method is the rail fence cipher. The ciphertext is created reading the pattern row by row. For example, to send the message "Meet me at the park" to Bob, Alice writes



She then creates the ciphertext "MEMATEAKETETHPR".

3.3.1 Continued

Example 3.23

Alice and Bob can agree on the number of columns.
Alice writes the same plaintext, row by row, in a table of four columns.

1	2	3	4
m	e	e	t
m	e	a	t
t	h	e	p
a	r	k	

She then creates the ciphertext "MMTAEEHREAEKTTP".

3.3.2 Keyed Transposition Ciphers

- **The keyless ciphers permute the characters**
  - by writing plaintext in one way and reading it in another way.
- **The permutation is done on the whole plaintext to create the whole ciphertext.**

- **Another method is to divide the plaintext into groups of predetermined size, called blocks, and then use a key to permute the characters in each block separately.**

3.3.2 Continued

Example 3.25

Alice needs to send the message "Enemy attacks tonight" to Bob..

e n e m y a t t a c k s t o n i g h t z

The key used for encryption and decryption is a permutation key, which shows how the character are permuted.

Encryption ↓

3	1	4	5	2
1	2	3	4	5

↑ Decryption

The permutation yields

E E M Y N T A A C T T K O N S H I T Z G

PLAINTEXT:

e n e m y a t t a c k s t o n i g h t z

key ⇨	***3***	***1***	***4***	***5***	***2***
	1	***2***	***3***	***4***	***5***
	e	***n***	***e***	***m***	***y***
	a	***t***	***t***	***a***	***c***
	k	***s***	***t***	***o***	***n***
	i	***g***	***h***	***t***	***z***

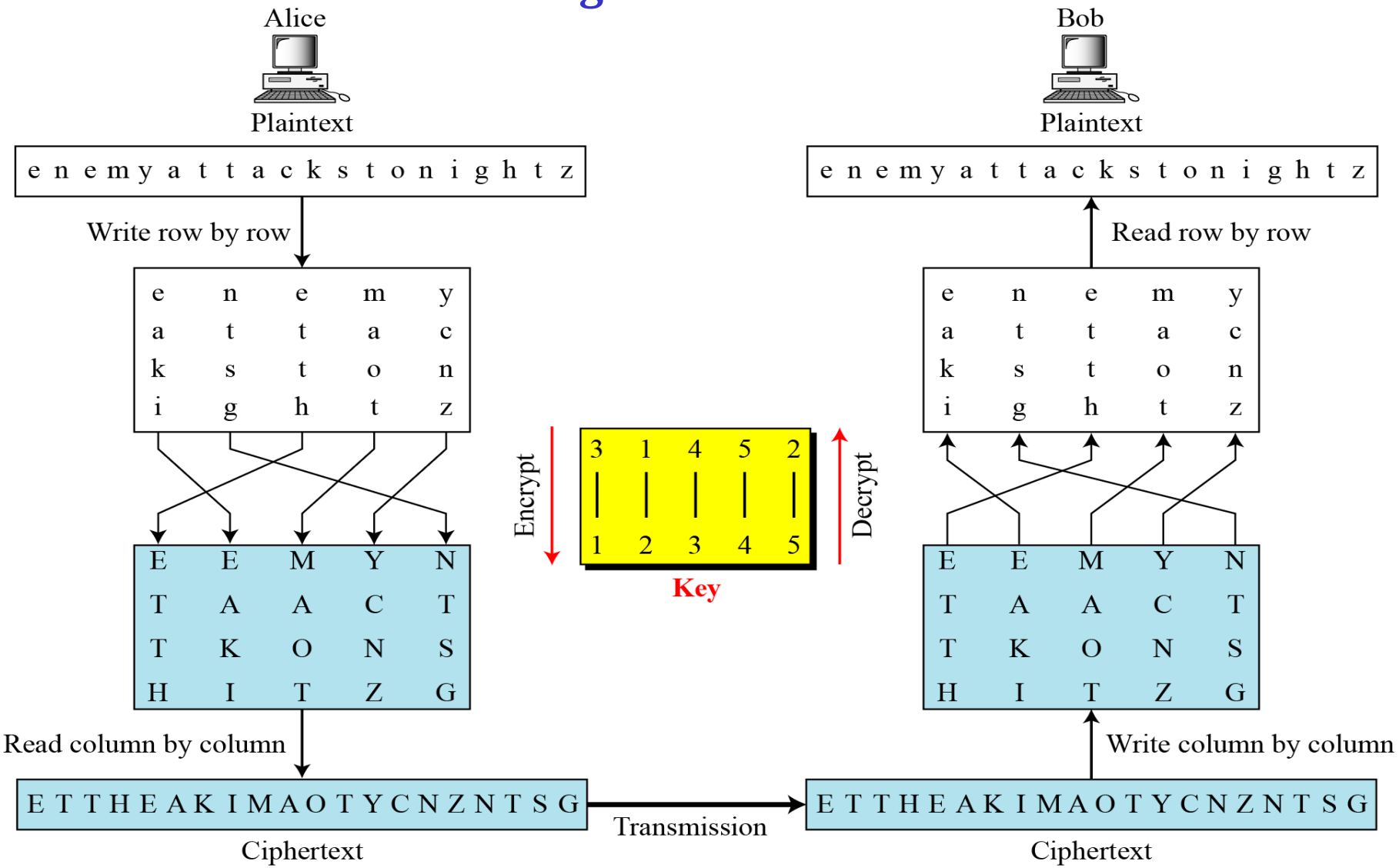
CIPHERTEXT:

E E M Y N T A A C T T K O N S H I T Z G

3.3.3 Combining Two Approaches

Example 3.26

Figure 3.21



Keys

In Example 3.27, a single key was used in two directions for the column exchange: downward for encryption, upward for decryption. It is customary to create two keys.

Figure 3.22 *Encryption/decryption keys in transpositional ciphers*

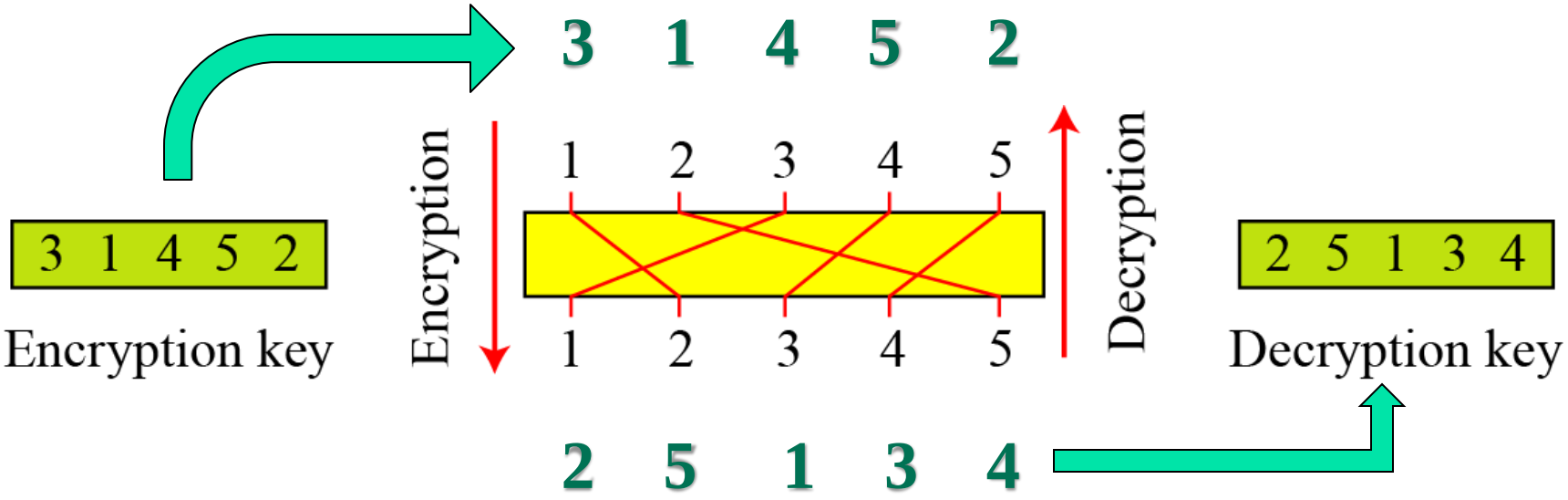
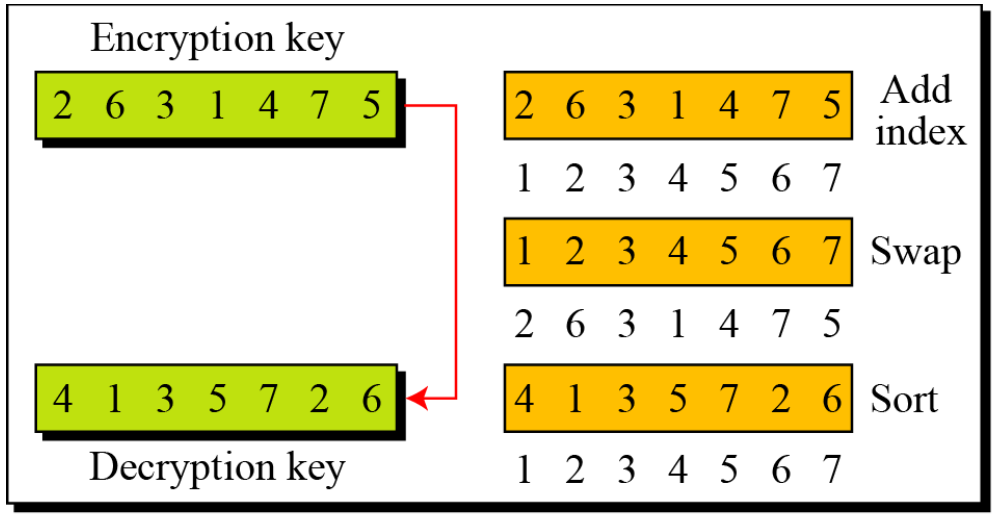


Figure 3.23 *Key inversion in a transposition cipher*

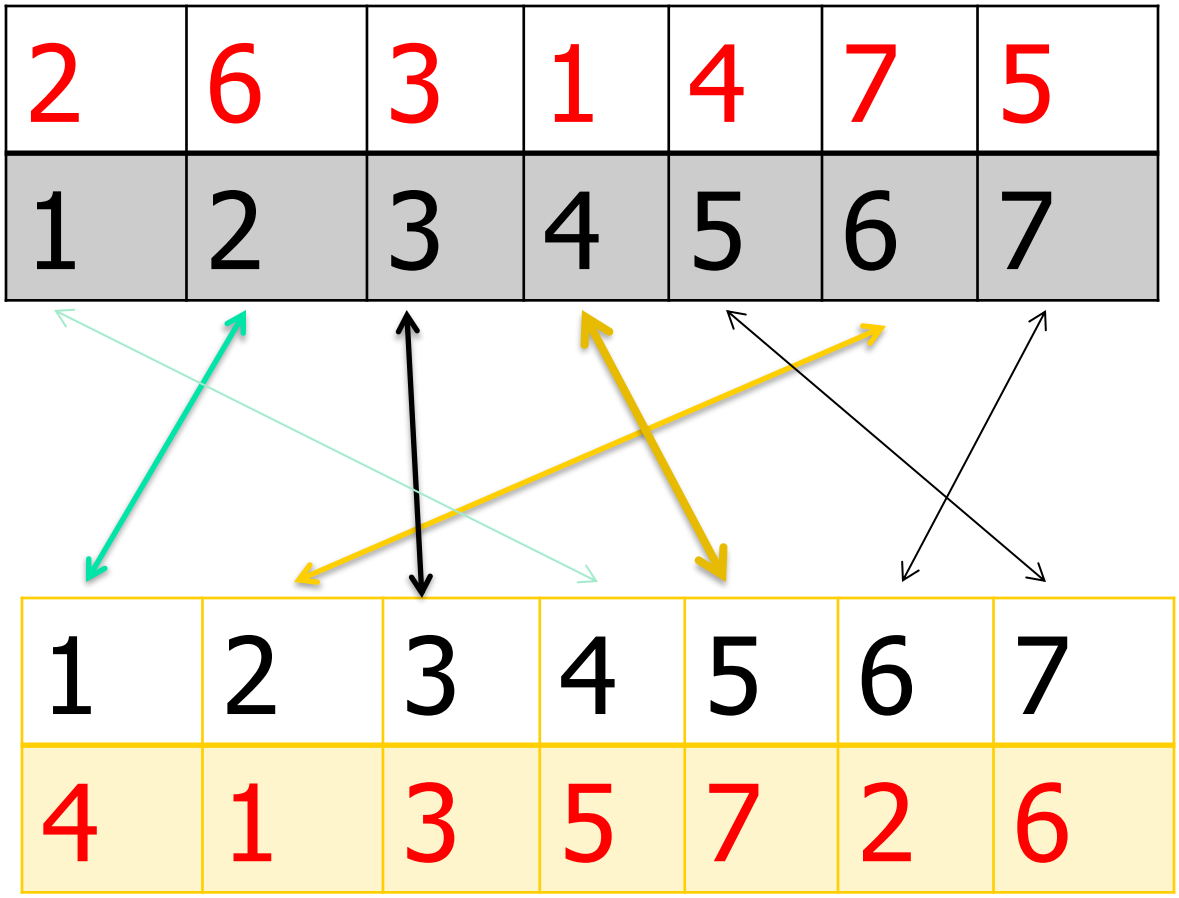


a. Manual process

```
Given: EncKey [index]
index ← 1
while (index ≤ Column)
  {
    DecKey[EncKey[index]] ← index
    index ← index + 1
  }
Return : DecKey [index]
```

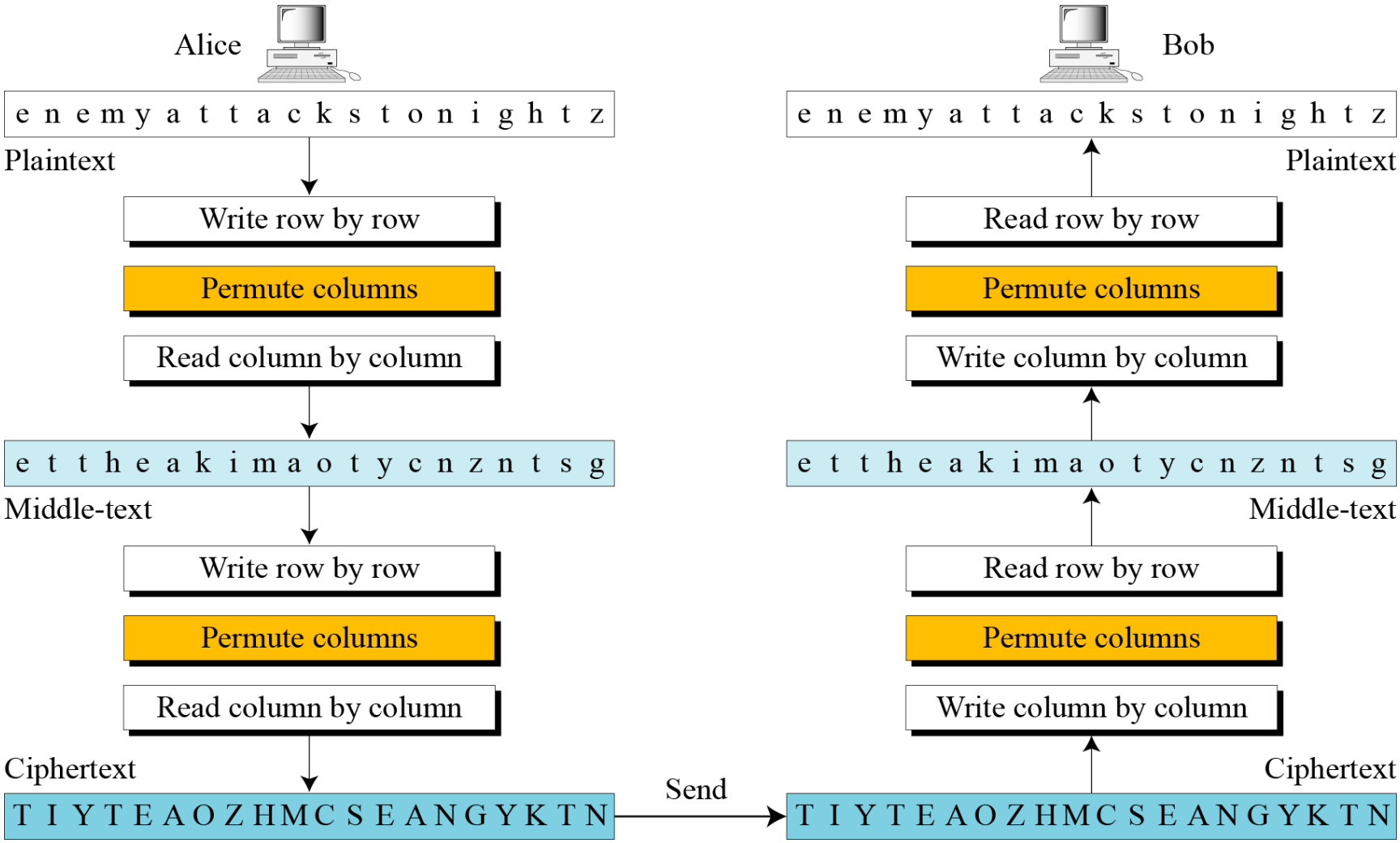
b. Algorithm

Key inversion in a transposition cipher



Double Transposition Ciphers

Figure 3.25 *Double transposition cipher*



Double Transposition Ciphers (Ex:)

3	1	4	5	2
e	n	e	m	y
a	t	t	a	c
k	s	t	o	n
i	g	h	t	z

3	1	4	5	2
e	e	m	y	n
t	a	a	c	t
t	k	o	n	s
h	i	t	z	g

CT1= ettheakimaotycnzntsg

3	1	4	5	2
e	t	t	h	e
a	k	i	m	a
o	t	y	c	n
z	n	t	s	g

3	1	4	5	2
t	e	h	e	t
i	a	m	a	k
y	o	c	n	t
t	z	s	g	n

CT2= tiyteaozhmcseangtktn

Product Ciphers

- *ciphers using substitutions or transpositions are not secure because of language characteristics*

- *hence consider using several ciphers in succession to make harder, but:*
  - *two substitutions make a more complex substitution*
  - *two transpositions make more complex transposition*
  - *but a substitution followed by a transposition makes a new much harder cipher*

- *this is bridge from classical to modern ciphers*

Modern Block Ciphers

- will now look at modern block ciphers
- provide secrecy and/or authentication services
- in particular will introduce DES (Data Encryption Standard)

Block vs. Stream Ciphers

- block ciphers process messages in into blocks, each of which is then en/decrypted
- like a substitution on very big characters
  - 64-bits or more
- stream ciphers process messages a bit or byte at a time when en/decrypting
- many current ciphers are block ciphers

Block Cipher Principles

- most symmetric block ciphers are based on a **Feistel Cipher Structure**
- block ciphers look like an extremely large substitution
- would need table of 2^{64} entries for a 64-bit block
- using idea of a product cipher

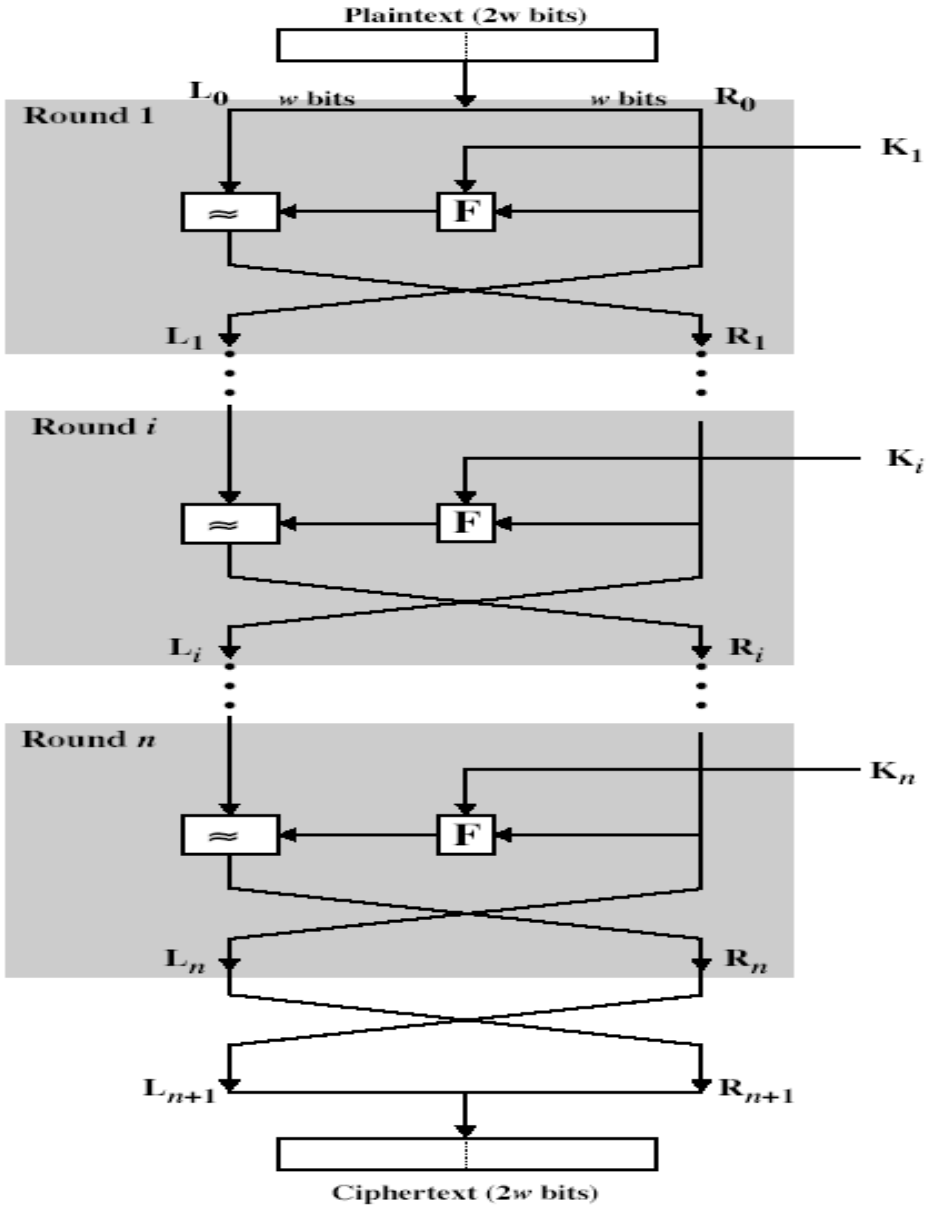
Claude Shannon and Substitution-Permutation Ciphers

- in 1949 Claude Shannon introduced idea of substitution-permutation (S-P) networks
  - modern substitution-transposition product cipher
- these form the basis of modern block ciphers
- S-P networks are based on the two primitive cryptographic operations we have seen before:
  - *substitution* (S-box)
  - *permutation* (P-box)
- provide *confusion* and *diffusion* of message

Feistel Cipher Structure

- Horst Feistel devised the **feistel cipher**
  - based on concept of invertible product cipher
- partitions input block into two halves
  - process through multiple rounds which
  - perform a substitution on left data half
  - based on round function of right half & subkey
  - then have permutation swapping halves
- implements Shannon's substitution-permutation network concept

Feistel Cipher Structure



Feistel Cipher Design Principles

- **block size**
  - increasing size improves security, but slows cipher
- **key size**
  - increasing size improves security, makes exhaustive key searching harder, but may slow cipher
- **number of rounds**
  - increasing number improves security, but slows cipher
- **subkey generation**
  - greater complexity can make analysis harder, but slows cipher
- **round function**
  - greater complexity can make analysis harder, but slows cipher
- **fast software en/decryption & ease of analysis**
  - are more recent concerns for practical use and testing

Feistel Cipher Decryption

